

HealthConnect Clinic

Week 6 Project Summary — Data Analytics Track

AnalystLab Africa Experience Lab | Prepared by: Moses Oluwatosin | Date: 10 September 2026

This summary is deliberately concise and complements — rather than repeats — the full Week 6 Advanced Analytics & Decision Support Report (docs/HealthConnect_Week6_Advanced_Analytics_Report.docx) and the executed notebook (notebooks/HealthConnect_Week6_Advanced_Analytics.ipynb).

1. What We Planned to Accomplish

Deepen and statistically validate the three strongest Week 5 findings (booking lead time, previous no-shows, reminder effectiveness), extend into segments Week 5 didn't formally test, refine the dashboard into a decision-support tool, and complete a real cross-track integration with the Data Science track — not repeat Week 5's EDA or rebuild the same dashboard.

2. What We Completed

- Chi-square tests and Cramer's V effect sizes for all three priority Week 5 findings.
- Two new formal segment tests not covered in Week 5: appointment type and reminder channel.
- Two interaction models (reminder × prior history, reminder × lead time) and one combined multivariate model.
- A refined 4-panel dashboard with statistical verdicts built into each panel.
- A ranked, evidence-backed feature-relevance table delivered as a concrete cross-track artefact.
- An updated assumptions/limitations/risks register reflecting what Week 6 resolved, refined, or newly found.

3. What We Improved From Week 5

- Moved from descriptive rate differences to statistically tested, effect-sized conclusions.
- Corrected the relative-importance picture: previous_no_shows looked dramatic by raw percentage but has a smaller effect size (0.089) than booking_lead_days (0.197).
- Downgraded appointment_type from "weak signal" to "not statistically supported" based on a formal test Week 5 never ran.
- Replaced the Week 5 dashboard's plain descriptive panels with decision-support panels that state their own statistical confidence.

4. What We Integrated

Delivered a concrete, reusable artefact (HealthConnect_Week6_Feature_Relevance.csv) to the Data Science track's Week 6 feature-refinement work — not just a conversation, but a file with p-values, effect sizes, and explicit include/exclude recommendations they can act on directly.

5. Track(s) Collaborated With

Data Science.

6. What Was Exchanged

Received: confirmation of their Week 5 baseline feature set. Provided: the ranked feature-relevance table plus an explicit caution not to engineer the reminder $\times$ prior-history interaction as a feature yet, since it isn't statistically supported.

7. What Changed As a Result

Data Science's feature-selection decisions for their Week 6 model refinement are now backed by statistical evidence rather than Week 5's descriptive rates alone, and one previously plausible feature idea (the interaction term) is now known to lack support before any engineering time is spent on it.

8. Key Findings / Development Outcomes

- Booking lead time, previous no-shows, reminder status, and reminder channel are all statistically significant ($p < 0.05$).
- Appointment type is not statistically significant ($p = 0.104$).
- The reminder $\times$ prior-history interaction is descriptively suggestive (2.65pp $\rightarrow$ 10.02pp reduction) but not statistically confirmed ($p = 0.143$).
- A combined multivariate model confirms lead time, prior history, and reminders each carry independent predictive signal.

9. Major Challenges

- Avoiding overstatement: the reminder $\times$ history interaction looked compelling descriptively but had to be reported honestly as unconfirmed once tested formally.
- Deciding how much new analysis to add without duplicating Week 5 — resolved by scoping strictly to the three chosen priority findings plus two genuinely untested segments.

10. Important Decisions

- Reported the non-significant interaction finding transparently rather than omitting it or overselling the descriptive pattern — this is more useful to the Data Science track than a falsely confident recommendation.
- Used effect size (Cramer's V), not just p-value, to rank findings by practical importance, since statistical significance alone doesn't indicate how much a variable actually matters.

11. Remaining Issues

- Sunday-appointment and missing-data items from Week 4/5 remain unresolved.
- The reminder $\times$ prior-history interaction needs re-testing with more data before being treated as confirmed.
- No causal test (e.g. an A/B reminder pilot) has been run — all findings remain observational.

12. Contribution to the Overall HealthConnect Project

This track has moved the shared project from "we think these patterns exist" (Week 5) to "these patterns are statistically confirmed, sized, and ready to inform both HealthConnect's operational decisions and the Data Science track's model features" (Week 6) — directly supporting the project's central goal of using data to reduce missed appointments.

13. Proposed Focus for Week 7

- Re-validate the three confirmed predictors once the Data Science track's refined model is available.
- Re-test the reminder $\times$ prior-history interaction if a larger sample becomes available.
- Formally test any new segment the Data Science track's error analysis surfaces.

Full documentation: [docs/HealthConnect_Week6_Advanced_Analytics_Report.docx](#) | Code: [notebooks/HealthConnect_Week6_Advanced_Analytics.ipynb](#) | Navigation: [README.md](#)